

Solution Requirements

Date	26 June 2026
Team ID	-
Project Name	EduGenie – AI-Powered Educational Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviour's, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviour's (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S. No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users register and log in securely with hashed passwords (bcrypt) and JWT-based session tokens.</i>	High
2	Authorization levels	<i>Protected API routes verify the JWT via auth middleware; each user only accesses their own data.</i>	High

3	Intelligent Q&A	<i>Users submit questions and receive markdown-formatted, structured answers from Google Gemini.</i>	High
4	Concept Explanation	<i>Users request an explanation of a topic at a chosen difficulty level and style (simple/detailed/examples/analogy)</i>	High
5	Quiz Generation	<i>Users generate MCQ, True/False, Fill-in-the-blank, or Short-answer quizzes on any topic with adjustable difficulty and count.</i>	High
6	Summarization	<i>Users summarize pasted text or uploaded PDF/DOCX/TXT files in short, detailed, bullet, takeaway, or key-term modes.</i>	High
7	Learning Roadmap	<i>Users generate a phased, milestone-based learning roadmap for a topic, level, and target duration.</i>	Medium
8	Smart Notes	<i>Users create, edit, delete, and organize notes into folders with autosave.</i>	Medium

9	Bookmarks	<i>Users bookmark useful AI responses under categories for later reference.</i>	Medium
10	AI History	<i>Every AI interaction is logged per-user and viewable in a history feed.</i>	Medium
11	Progress Analytics	<i>The system tracks streaks, focus score, sessions completed, weekly goals, and quiz scores per user.</i>	Medium
12	Settings	<i>Users update theme (light/dark), AI model preference, notification and weekly-goal settings.</i>	Low

Step 2: Non-Functional Requirements (NFR)

S. No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance	Fully asynchronous Fast API backend with non-blocking MongoDB (Motor) operations for fast API responses.	Average API response time < 500 MS for normal requests; page load time < 2 seconds.
2	Scalability	Modular router/service architecture (auth, ai, notes, bookmarks, history, progress, settings) allows independent scaling	Support 1,000+ concurrent users with minimal performance degradation; new modules can be added without affecting existing services.

3	Security	Environment-variable managed secrets, bcrypt password hashing, JWT auth, and configurable CORS.	Passwords stored only as bcrypt hashes; JWT-protected endpoints; no hard-coded secrets; HTTPS supported.
4	Usability	Responsive frontend with light/dark themes and a consistent design system across all pages.	Responsive on mobile, tablet, and desktop; common tasks completed within 3 clicks; consistent UI throughout the application.
5	Reliability	Global exception handler returns consistent error responses; MongoDB Atlas provides managed durability.	System uptime $\geq 99\%$; graceful error handling; no application crashes during normal usage.
6	Maintainability	Clean separation of concerns (routers / services / schemas / utils) for easy extension.	Well-structured, modular code with reusable components; new features can be integrated with minimal code changes.
7	Portability	Deployable on Render, Railway, Fly.io or any VPS for the backend and Vercel/Netlify for the frontend.	Application deploys successfully across supported platforms without source code modifications.